diamond (an important system for quantum sensing discussed in Chap. 6). The solution to this complicated potential can only be found using sophisticated numerical methods, which yield approximate solutions. Nevertheless, there are a few simple potentials which both describe realistic physical systems very well and are amenable to analytical solution. One example is the Coulomb potential, which accurately describes the potential experienced by an electron in a hydrogen atom. Another important example is the harmonic potential, which we will now discuss.

3.1.2 Harmonic potential

A harmonic potential with angular frequency ω for a particle of mass m moving in one dimension is given by

$$V(x) = \frac{1}{2}m\omega^2 x^2, \tag{3.14}$$

and the Hamiltonian (3.2) is $\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2$. It is often convenient to rescale the position and momentum operators,

$$\hat{x} \to \sqrt{\frac{m\omega}{\hbar}}\,\hat{x} \quad \text{and} \quad \hat{p} \to \frac{1}{\sqrt{m\omega\hbar}}\,\hat{p}. \tag{3.15}$$

The commutator of the rescaled operators is then

$$[\hat{x}, \hat{p}] = i, \tag{3.16}$$

and the Hamiltonian takes the form

$$\hat{H} = \frac{1}{2}\hbar\omega(\hat{x}^2 + \hat{p}^2). \tag{3.17}$$

This harmonic oscillator Hamiltonian can represent a large number of different physical systems used in quantum information technology. They include, for example, the electromagnetic field (particularly optical waves and microwaves), the inductor-capacitor electronic circuit, acoustic sound waves in bulk materials, and mechanical oscillators such as oscillating membranes or cantilevers. Despite the apparent differences in these systems, they can all be described by the same Hamiltonian and thus the same mathematics. However, the'position' and 'momentum' operators $\hat{x}$, $\hat{p}$ represent different quantities depending on the problem. For mechanical oscillators, they represent actual position and momentum, but for the electromagnetic field, $\hat{x}$ and $\hat{p}$ represent the amplitude and phase quadratures of the field (as will be discussed below), while for inductor-capacitor electronic circuits, they represent charge and current.

To find the energy eigenvalues and eigenstates for the quantum harmonic oscillator, we must solve the time-independent Schrödinger equation. However, for this system there exists a much more elegant approach than solving differential equation for the wavefunction directly. We start by introducing the operators

$$\hat{a} = \frac{1}{\sqrt{2}}(\hat{x} + i\hat{p}) \quad \text{and} \quad \hat{a}^\dagger = \frac{1}{\sqrt{2}}(\hat{x} - i\hat{p}). \tag{3.18}$$

These operators are called the *creation* and *annihilation* operators, respectively, and are referred to jointly as *ladder operators*, for reasons that will become clear shortly. From the canonical commutation relation $[\hat{x}, \hat{p}] = i$, it follows that

$$[\hat{a}, \hat{a}^\dagger] = 1. \tag{3.19}$$

The ladder operators are not Hermitian and thus not observables, and therefore they do not correspond to physically observable parameters of any system. Nevertheless, their importance should not be underestimated! They enable many algebraic calculations, and moreover, observable quantities can be formed through sums and products of them. In particular, we have

$$\hat{x} = \frac{1}{\sqrt{2}}(\hat{a} + \hat{a}^\dagger), \tag{3.20}$$

$$\hat{p} = \frac{1}{i\sqrt{2}}(\hat{a} - \hat{a}^\dagger). \tag{3.21}$$

Using the commutation relations, the harmonic oscillator Hamiltonian can then be expressed as

$$\hat{H} = \hbar\omega(\hat{a}^\dagger\hat{a} + \frac{1}{2}) = \hbar\omega(\hat{n} + \frac{1}{2}), \tag{3.22}$$

where we have introduced a new Hermitian operator $\hat{n} = \hat{a}^\dagger\hat{a}$ known as the *number operator*. Clearly, if we can find the eigenvalues and eigenstates of $\hat{n}$ then we also determine the energy eigenvalues and eigenstates. Thus, we need to solve the eigenequation

$$\hat{n}|n\rangle = n|n\rangle. \tag{3.23}$$

A priory, the eigenvalues could be any numbers, but the suggestive notion here is of course not a coincidence – we will show that the n's are non-negative integers.

First, note that for a normalised eigenstate $|n\rangle$ of $\hat{n}$ with eigenvalue n, we have that

$$n = \langle n|\hat{n}|n\rangle = \langle n|\hat{a}^\dagger\hat{a}|n\rangle \geq 0, \tag{3.24}$$

because the last expression is the norm of the state $\hat{a}|n\rangle$. Thus, the eigenvalues of $\hat{n}$ cannot be negative. Next, we observe that

$$\hat{n}\hat{a} = \hat{a}^\dagger\hat{a}\hat{a} = (\hat{a}\hat{a}^\dagger - [\hat{a}, \hat{a}^\dagger])\hat{a} = \hat{a}(\hat{n} - 1), \tag{3.25}$$

$$\hat{n}\hat{a}^\dagger = \hat{a}^\dagger\hat{a}\hat{a}^\dagger = \hat{a}^\dagger(\hat{a}^\dagger\hat{a} + [\hat{a}, \hat{a}^\dagger]) = \hat{a}^\dagger(\hat{n} + 1). \tag{3.26}$$

This means that, for any eigenstate $|n\rangle$ we have

$$\hat{n}\hat{a}|n\rangle = (n - 1)\hat{a}|n\rangle, \tag{3.27}$$

$$\hat{n}\hat{a}^\dagger|n\rangle = (n + 1)\hat{a}^\dagger|n\rangle. \tag{3.28}$$

Thus, $\hat{a}|n\rangle$ and $\hat{a}^\dagger|n\rangle$ are also eigenstates of $\hat{n}$ with eigenvalues $n-1$ and $n+1$, respectively. If we keep applying $\hat{a}$ repeatedly, we generate states with smaller and smaller eigenvalues, $n \to n - 1 \to n - 2 \to \ldots$. Eventually, we will get an eigenstate with negative eigenvalue, in contradiction with Eq. (3.24). That is, unless n is an integer, in which case $\hat{a}^n|n\rangle$ is an eigenstate with eigenvalue zero. In this case $\hat{a}^{n+1}|n\rangle = 0$ from Eq. (3.25) and further applications of $\hat{a}$ has no effect. We conclude that n must be a non-negative integer. The smallest